Daily Expense Management System

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Abstract
Many people find it difficult to track daily expenses effectively, resulting in poor budgeting and financial
To address this, I developed a Daily Expense Management System to help users record, categorize, ar
enhancing financial visibility and supporting more efficient budgeting decisions.

## 1. Project Overview

Many existing financial tools have complex interfaces, lack real-time feedback, and weak categorizatior

- Simple expense recording
- Automatic classification
- Real-time reports and charts
- Alerts for budget control
- Cloud backup and data security

## 2. Literature Review

- Gupta et al. (2018) emphasized that user-friendly interfaces and visual charts are key to higher adopti
- Chen & Liang (2020) introduced an NLP-based classification method that significantly improved accur
- Singh et al. (2021) highlighted the effectiveness of visual dashboards and budget tracking on user beh
- Kumar & Ray (2019) focused on the importance of encryption and cloud storage for financial data [4].

This project integrates these insights to improve classification, visualization, and security.

## 5. References

[1] Gupta, R. et al. (2018). Mobile-Based Personal Finance Management System. IEEE.
[2] Chen, Y. & Liang, J. (2020). Expense Categorization Using NLP. Elsevier.
[3] Singh, A. et al. (2021). Budget Monitoring Dashboards. Springer.
[4] Kumar, N. & Ray, S. (2019). Data Security in Finance Apps. IEEE.

## 3. Methodology

This section outlines the technical approach used in developing the Daily Expense Management System, including tools, structure, and implementation strategies.

3.1 Development Tools and Environment

The project was developed in Python 3.10, using Visual Studio Code as the primary IDE. Version control was managed using GitHub. The program runs on standard Windows and macOS machines without requiring special hardware.

3.2 Data Storage and Management

All expense data is stored in a local SQLite3 database, which is lightweight and easy to integrate with Python. Tables were created for user accounts, transactions, categories, and budget limits. The database enables CRUD operations, data persistence, and category-based filtering.

Sample schema snippet:

```
CREATE TABLE transactions (
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   date TEXT,
   category TEXT,
   amount REAL,
   note TEXT
);
```

3.3 Program Features and Code Structure

The project uses a modular Python structure with clearly separated files for database operations, interface logic, and chart rendering. Main features include:

- Adding and editing expense records
- Auto-categorization based on keywords
- Generating monthly reports and pie charts
- Budget threshold alerts

Charting is implemented using matplotlib, and the user interface uses Tkinter.

3.4 Security and Backup

The database can be exported for cloud backup, and user data is stored locally with optional encryption. Future versions may integrate Firebase or Google Drive for real-time syncing.